MATH 565 Monte Carlo Methods

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Instructions:

i. *This test has **THREE questions**. Attempt all. The maximum number of points is **100**.*

ii. *The time allowed is 75 minutes.*

iii. *This test is closed book, but you may use **4** double-sided letter-size sheets of notes.*

iv. *No calculators or other electronic devices are allowed. Phones must be placed in your bags under your desks or at the front of the room. Hands must be on top of your desks.*

v. *You are expected to simplify any expressions that can be simplified as whole numbers, e.g.,* $36/15 = 12/5$*, but you may leave other expressions as is, e.g.,* $\sqrt{47}$ *or* $18/29$*.*

vi. *Show all your work to justify your answers. Answers without adequate justification will not receive credit.*

vii. *Print your name clearly on each answer sheet.*

viii. *Off-site students may contact the instructor as directed by your syllabus.*

Q	Max	Score
1	36	
2	40	
3	24	
Tot	**100**	

ChatGPT/Codex assisted in the construction of this exam.

I understand these instructions and have not relied on any help for this exam beyond what is allowed, nor provided any help to anyone else beyond what is allowed.

_____ _____

Signature *Date*

_____

Printed Name

1. (36 points) Let $Y = f(\boldsymbol{X})$, $\boldsymbol{X} \sim \varrho_{\text{tar}}$, and $\mu = \mathbb{E}(Y)$. Assume the required moments exist and proposal support conditions hold. Explain your answers.

   a. (12 points) For $\boldsymbol{Z} \sim \varrho_{\text{prop}}$, define $Y_{\text{IS}} = f(\boldsymbol{Z})\varrho_{\text{tar}}(\boldsymbol{Z})/\varrho_{\text{prop}}(\boldsymbol{Z})$. Do Y and Y_{IS} have the same mean? The same variance?

   Answer: They have the same mean:

$$\mathbb{E}(Y_{\text{IS}}) = \int f(\boldsymbol{z})\varrho_{\text{tar}}(\boldsymbol{z})\,\mathrm{d}\boldsymbol{z} = \mu.$$

   Their variances need not agree. For example, take target density $2x$ *on* $(0,1)$*, a uniform proposal, and* $f(x) = x$*. Then* $\operatorname{var}(Y_{\text{IS}}) = \operatorname{var}(2Z^2) = 4/5 - 4/9 = 16/45$*, whereas* $\operatorname{var}(Y) = 1/2 - 4/9 = 1/18$*.*

b. (12 points) A control $g(\boldsymbol{X})$ has known nonzero mean $\mathbb{E}[g(\boldsymbol{X})] = \nu$ and positive variance, $\operatorname{var}\big(g(\boldsymbol{X})\big) = \sigma_g^2$.

i. Define $Y_{\mathrm{CV}} = Y - g(\boldsymbol{X})$. Is $\mathbb{E}(Y_{\mathrm{CV}}) = \mu$?

ii. Construct an unbiased, variance-minimizing control-variate estimator of μ using the control $g(\boldsymbol{X})$, and determine the coefficient.

Answer:

i. No. Since the control has nonzero mean,

$$\mathbb{E}(Y_{\mathrm{CV}}) = \mathbb{E}[Y - g(\boldsymbol{X})] = \mu - \nu \neq \mu.$$

ii. Center the control and introduce a coefficient β:

$$Y_\beta = f(\boldsymbol{X}) - \beta\{g(\boldsymbol{X}) - \nu\}, \qquad \mathbb{E}(Y_\beta) = \mu$$

for every fixed β. For IID $\boldsymbol{X}_i \sim \varrho_{\mathrm{tar}}$, use

$$\widehat{\mu}_{\mathrm{CV}} = \frac{1}{n}\sum_{i=1}^{n}\big[f(\boldsymbol{X}_i) - \beta\{g(\boldsymbol{X}_i) - \nu\}\big].$$

This estimator is unbiased, with variance

$$\operatorname{var}(\widehat{\mu}_{\mathrm{CV}}) = \frac{1}{n}\big[\operatorname{var}(Y) - 2\beta \operatorname{cov}\{Y, g(\boldsymbol{X})\} + \beta^2\sigma_g^2\big].$$

Differentiating with respect to β gives $2[\beta\sigma_g^2 - \operatorname{cov}\{Y, g(\boldsymbol{X})\}]/n$. Since $\sigma_g^2 > 0$, the unique minimizing coefficient is

$$\beta_* = \frac{\operatorname{cov}\{Y, g(\boldsymbol{X})\}}{\sigma_g^2}.$$

The minimum variance is

$$\operatorname{var}(\widehat{\mu}_{\mathrm{CV}}) = \frac{1}{n}\left[\operatorname{var}(Y) - \frac{\operatorname{cov}\{Y, g(\boldsymbol{X})\}^2}{\sigma_g^2}\right].$$

If the coefficient is estimated, an independent pilot preserves unbiasedness; fitting it to the same observations need not do so. The population coefficient β_* gives the stated minimum.

c. (12 points) Let $\boldsymbol{u}_0^{(1)}, \boldsymbol{u}_1^{(1)}, \ldots$ and $\boldsymbol{u}_0^{(2)}, \boldsymbol{u}_1^{(2)}, \ldots$ be two copies of the same digital sequence (e.g., the Sobol' sequence), with independent uniform random digital shifts. Suppose $\boldsymbol{T}(\boldsymbol{U}) \sim \varrho_{\text{tar}}$ for $\boldsymbol{U} \sim \text{Unif}[0,1]^d$.

i. What is $\mathbb{E}[f(\boldsymbol{T}(\boldsymbol{u}_1^{(1)}))]$?

ii. Are $f(\boldsymbol{T}(\boldsymbol{u}_1^{(1)}))$ and $f(\boldsymbol{T}(\boldsymbol{u}_3^{(2)}))$ independent?

iii. Are $f(\boldsymbol{T}(\boldsymbol{u}_1^{(1)}))$ and $f(\boldsymbol{T}(\boldsymbol{u}_3^{(1)}))$ independent?

Answer:

i. A uniform random digital shift makes each point marginally uniform. Thus

$$\mathbb{E}[f(\boldsymbol{T}(\boldsymbol{u}_1^{(1)}))] = \int_{[0,1]^d} f(\boldsymbol{T}(\boldsymbol{u}))\,\mathrm{d}\boldsymbol{u} = \mu.$$

ii. Yes. Each value is a function of one of the two independent shifts.

iii. Not necessarily. Both points use the same shift. For example, in base 2 and one dimension with $\boldsymbol{T}$ the identity and $f(x) = x$, the two shifted points are related by a fixed digital XOR, so one determines the other; they are nondegenerate and dependent.

2. (40 points) Consider a Metropolis–Hastings independence sampler on $\{0,1,2\}$ with unnormalized target and proposal masses

x	0	1	2
$\widetilde{\varrho}_{\text{tar}}(x)$	1	2	3
$\widetilde{\varrho}_{\text{prop}}(x)$	1	1	1

Each proposal $Z_i \sim \varrho_{\text{prop}}$ is independent of the current state and previous draws. The decision uniforms $V_i \sim \text{Unif}[0,1]$ are mutually independent and independent of all proposals. Let $\mu = \mathbb{E}_{\text{tar}}(X)$. Justify your answers.

a. (15 points) Give the probability $\alpha(x,z)$ of accepting proposal z from state x.

Answer: Write $\varrho_{\text{tar}} = \widetilde{\varrho}_{\text{tar}}/C_{\text{tar}}$ and $\varrho_{\text{prop}} = \widetilde{\varrho}_{\text{prop}}/C_{\text{prop}}$. For independent proposals,

$$\begin{aligned}
\alpha(x,z) &= \min\left\{1, \frac{\varrho_{\text{tar}}(z)\varrho_{\text{prop}}(x)}{\varrho_{\text{tar}}(x)\varrho_{\text{prop}}(z)}\right\} \\
&= \min\left\{1, \frac{[\widetilde{\varrho}_{\text{tar}}(z)/C_{\text{tar}}][\widetilde{\varrho}_{\text{prop}}(x)/C_{\text{prop}}]}{[\widetilde{\varrho}_{\text{tar}}(x)/C_{\text{tar}}][\widetilde{\varrho}_{\text{prop}}(z)/C_{\text{prop}}]}\right\} \\
&= \min\left\{1, \frac{\widetilde{\varrho}_{\text{tar}}(z)\widetilde{\varrho}_{\text{prop}}(x)}{\widetilde{\varrho}_{\text{tar}}(x)\widetilde{\varrho}_{\text{prop}}(z)}\right\}.
\end{aligned}$$

The numerator and denominator contain the same factor $1/(C_{\text{tar}}C_{\text{prop}})$, which cancels. The target normalizer C_{tar} would cancel in general Metropolis–Hastings, even with state-dependent proposals. Here the proposal normalizer C_{prop} also cancels because this is an independence sampler: the proposal distribution is the same from every current state, so C_{prop} depends on neither x nor z. For a general state-dependent proposal, the forward and reverse normalizers may differ and need not cancel. Independence alone does not make

the forward and reverse probabilities equal. Here the proposal is uniform, with $\widetilde{\varrho}_{\text{prop}}(x) = \widetilde{\varrho}_{\text{prop}}(z) = 1$, *so*

$$\alpha(x, z) = \min\left\{1, \frac{\widetilde{\varrho}_{\text{tar}}(z)}{\widetilde{\varrho}_{\text{tar}}(x)}\right\} = \min\left\{1, \frac{z+1}{x+1}\right\}.$$

b. (5 points) How does proposal independence allow the use of an unnormalized proposal in $\alpha(x, z)$?

Answer: The proposal distribution is the same from every current state:

$$\Pr(Z_i = z \mid X_i = x) = \frac{\widetilde{\varrho}_{\text{prop}}(z)}{C_{\text{prop}}}.$$

The reverse proposal probability is $\widetilde{\varrho}_{\text{prop}}(x)/C_{\text{prop}}$, *with the same constant. Hence*

$$\frac{\varrho_{\text{prop}}(x)}{\varrho_{\text{prop}}(z)} = \frac{\widetilde{\varrho}_{\text{prop}}(x)/C_{\text{prop}}}{\widetilde{\varrho}_{\text{prop}}(z)/C_{\text{prop}}} = \frac{\widetilde{\varrho}_{\text{prop}}(x)}{\widetilde{\varrho}_{\text{prop}}(z)}.$$

For a state-dependent proposal, the forward and reverse normalizers may differ and need not cancel. Independence guarantees a common normalizer, though other proposals can also have this property. We still need to sample from the actual proposal distribution.

c. (20 points) Starting at $X_0 = 1$, tabulate $\alpha(X_i, Z_i)$ and X_{i+1}, then estimate μ using $X_1, \ldots, X_4$.

i	0	1	2	3
proposal Z_i	2	0	1	0
decision V_i	3/4	3/4	1/2	1/4

Answer: Accept when $V_i \le \alpha(X_i, Z_i)$; *otherwise retain* X_i:

i	X_i	Z_i	$\alpha(X_i, Z_i)$	*accepted?*	X_{i+1}
0	1	2	1	*yes*	2
1	2	0	1/3	*no*	2
2	2	1	2/3	*yes*	1
3	1	0	1/2	*yes*	0

Thus $\widehat{\mu}_{\text{MCMC}} = (2 + 2 + 1 + 0)/4 = 5/4$. *Keep repeated states: they record the time the chain spends at each state. Omitting them generally changes the sampled distribution.*

This problem should be familiar if you understand your homework.

3. (24 points) Generate the four points with indices $i = 0, 1, 3, 7$ of the two-dimensional shifted lattice sequence with base-2 van der Corput ordering, generator $\boldsymbol{\zeta} = (1, 5)$, and shift $(1/4, 5/8)$, added modulo 1.

Answer: Using $\boldsymbol{u}_i = [\phi_2(i)(1, 5) + (1/4, 5/8)] \bmod 1$*, where reduction is modulo 1, componentwise, gives*

i	$\phi_2(i)$	*unshifted point*	$\boldsymbol{u}_i$
0	0	$(0, 0)$	$(1/4, 5/8)$
1	1/2	$(1/2, 1/2)$	$(3/4, 1/8)$
3	3/4	$(3/4, 3/4)$	$(0, 3/8)$
7	7/8	$(7/8, 3/8)$	$(1/8, 0)$

Test 2 Scores (out of 100)
Number of Students: 19, Minimum: 0, Maximum: 100, Mean: 58.7, Median: 63
Standard Deviation: 27.2, Quartiles (Q1, Q3): (35, 82)

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10|0
 9|1
 8|299
 7|147
 6|37
 5|15
 4|34
 3|25
 2|58
 1|
 0|0
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